

Atharva Ajay Wani, MS

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Education

Master of Science in Robotics and Autonomous Systems: Mechanical and Aerospace Engineering	May 2024
• Arizona State University, Tempe, AZ	GPA: 3.78/4.00
• Relevant Coursework: Mechatronics device innovation, Programming of IoT devices, Machine Learning, Wearables.	
Bachelor of Technology Mechanical Engineering: Machine Design minor	Aug 2021
• Manipal Academy for Higher Education, Manipal, Karnataka, India.	GPA: 3.53/4.00

Skills

CAD/CAE	Siemens NX, SolidWorks, Catia V5 & 3DX, Fusion360, Creo, AutoCAD, ANSYS.
Fabrication/Prototyping	3D printing (FDM/SLA/SLS), CNC Lathe/Mill, GD&T, Laser Cutter, electric circuits design.
Computer Skills	Python, ROS2, Linux, Gazebo, MATLAB/Simulink, XML, Git, Arduino, ESP32, C++, OpenCV.

Professional Experience

Mechanical Engineer III, ASML US, LP	Nov 2024 - Present
- Designed and developed complex precision machine systems for semiconductor manufacturing equipment. - Collaborated with cross functional teams to optimize designs for performance, reliability and cost using Siemens NX. - Produced test plans, prototypes, and analyses for mechanisms used for sub-micron level precision positioning of DUV and EUV photomasks for semiconductor manufacturing. - Worked on supplier selection, qualification, and design transfer to new supplier to realize a 5-15% reduction in part cost.	
Teaching Assistant, Massachusetts Institute of Technology Lincoln Laboratory	June 2024 - August 2024
- Developed coursework on Microcontrollers, Sensors, Electro-mechanical Actuators, improving instructional quality. - Organized a hackathon for 30+ participants, fostering innovation and practical skills. - Mentored high school students in solving electrical design challenges in the BWSI Microelectronics Course, improving their understanding and learning outcomes.	
Tech trainee intern, Tesla	Aug 2023 - May 2024
- Analyzed system communications, performance using advanced diagnostic tools - CAN analyzer and Pico scope. - Reduced diagnostic time by 30% through successful Root Cause Analysis and resolution of hardware, software issues. - Applied expertise in electric powertrain design principles and autonomous driving system interactions.	
Mechanical Design Engineer Intern, ACME process systems pvt. Ltd.	Feb 2021 - Aug 2021
- Integrated ANSYS for fluid flow simulation in R&D, resulting in a 30% increase in simulation accuracy. - Converted 10 chemical mixer designs into optimized 3D models for CFD, improving product design efficiency by. - Established streamlined procedures for 3D modeling, fluid dynamics simulation setup, reducing analysis time by 20%.	
Vehicle dynamics engineer, Team Manipal racing	Aug 2018 - May 2019
- Collaborated in a cross-functional team of 35 students to design a BAJA-SAE vehicle. - Utilized iterative design optimization techniques to develop 20% lighter steering knuckles. - Played a key role in the fabrication, assembly, and testing of 2 key vehicle components, Steering knuckles, and Suspension linkages, hands-on experience and proficiency in manufacturing processes, and using machine tools.	

Projects

Neurological Disability Assistive Technology Wearable Medical Device	Jan 2023 - Nov 2024
Mechanical Engineer, Arizona State University - Led a team of 6 students collaborating with doctors at Barrow Neurological Institute, optimizing fitment and ergonomics of electric stimulation pads. Oversaw complete Project Management and Technical Development. - Spearheaded electro-mechanical design, Rapid-prototyping, 3D-Printing responsibilities methods to 3D print electric stimulation pad design prototypes, enhancing user accessibility and ease of operation. - Manufactured and assembled the innovative prototype using laser cutting, 3D Printing FDM/SLS with medical grade materials resulting in Best-Project Award and patent application. (US Patent application number: 66/611,833).	
Collision avoidance system for Autonomous vehicle in ethically challenging situation	Jan 2024 - May 2024
- Identified autonomous driving scenario that may occur 10-15% more often as number of autonomous vehicles increase. - Designed MATLAB/Simulink simulations for decision making and collision avoidance algorithms. - Used Automated Driving toolbox, and Model Predictive Control toolbox.	
Prosthetic Hand	Jan 2023 - May 2024
Designed a realistic 3D printed prosthetic hand with real-time motor control system, sensor fusion integration.	
Wildlife camera trap with live stream on mobile app. (ESP-32, Arduino, NodeJS, AWS, HTML, CSS, git)	Aug 2023 - Dec 2023
Additional Skills - Collaborated with doctors in clinical requirements translation, disposable medical equipment design (ASU). - Hands on experience with fabrication, test equipment, test and validation processes and data analysis (ASU). - Debugging and analyzing complex electrical circuits in complex autonomous systems (Tesla).	